Markov Chains

Sequences, transition matrices, and long-run behaviour

Prof. Nipun Batra

Mathematical Foundations for AI · IIT Gandhinagar

Tomorrow depends on today

A two-state weather model

Each day is either sunny S or rainy R.

Assume:

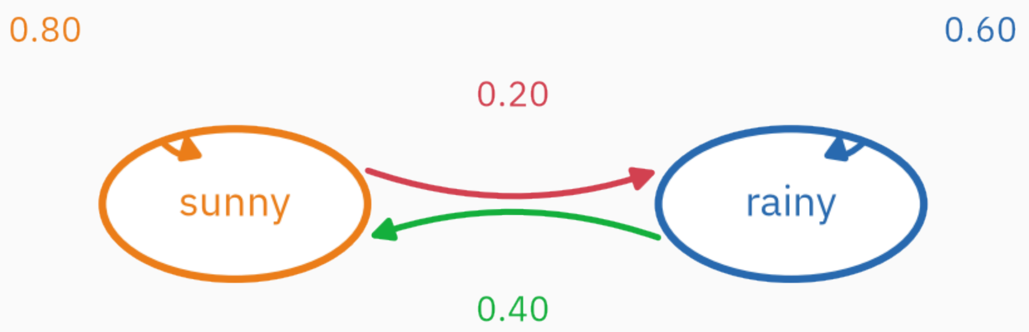
$$P(S_{t+1}|S_t) = 0.8, \quad P(R_{t+1}|S_t) = 0.2,$$

$$P(S_{t+1}|R_t) = 0.4, \quad P(R_{t+1}|R_t) = 0.6.$$

Tomorrow is random, but today's state changes its probabilities.

State diagram

V VISUAL



rows: current state; arrows: next-state probabilities

Outgoing probabilities from each state sum to one.

The Markov assumption

A first-order Markov model assumes

$$\boxed{P(X_{t+1}|X_t, X_{t-1}, ..., X_1) = P(X_{t+1}|X_t).}$$

Given the present state, earlier states add no further information about the next state.

This is a modeling assumption, not a universal property of sequences.

What the assumption removes

Without a Markov assumption, predicting X_{t+1} may require a table for every possible history.

For K states, a length-t history has K^t possibilities.

A first-order homogeneous chain uses only a $K \times K$ transition table, shared across time.

The reduction is substantial; the lost memory may also matter.

Examples and non-examples

Reasonable first approximations:

1. weather category tomorrow from weather today,
2. a board-game position after one move,
3. user state after one interaction,
4. next character from the current character.

Often insufficient:

1. seasonal weather without the month,
2. language when only one previous character is kept,
3. medical history summarized by an incomplete state.

Learning outcomes

By the end of this lecture you will be able to:

1. factor a sequence probability using the Markov assumption,
2. represent a chain by a row-stochastic transition matrix,
3. propagate a state distribution by matrix multiplication,
4. compute multi-step transition probabilities,
5. derive ⭐ a stationary distribution by equations and eigenvectors,
6. distinguish irreducibility from aperiodicity,
7. run power iteration,
8. and derive PageRank as a damped random walk on a web graph.

The entire chain is a matrix

Fix an orientation

Order states as (S, R). Put current state in rows and next state in columns:

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

Entry P_{ij} means

$$P_{ij} = P(X_{t+1} = j | X_t = i).$$

Every row sums to one and every entry is non-negative. Such a matrix is **row-stochastic**.

Read the matrix both ways

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

Row view:

1. if today is sunny, tomorrow is $(0.8, 0.2)$ over (S, R),
2. if today is rainy, tomorrow is $(0.4, 0.6)$.

Column view:

1. the sunny column collects all ways to arrive at sunny,
2. the rainy column collects all ways to arrive at rainy.

A distribution over states

Let

$$\boldsymbol{\mu}_t = (P(X_t = S), P(X_t = R))$$

be a row vector.

The next distribution is

$$\boxed{\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t P.}$$

This is the law of total probability written as matrix multiplication.

One-day prediction

Suppose today is certainly sunny:

$\boldsymbol{\mu}_0 = (1, 0).$

$$\boldsymbol{\mu}_1 = (1, 0)\begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} = (0.8, 0.2).$$

The first row of P is the next-day distribution.

Two-day prediction

$$\boldsymbol{\mu}_2 = \boldsymbol{\mu}_1 P = (0.8, 0.2)\begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

$$P(X_2 = S) = 0.8(0.8) + 0.2(0.4) = 0.72.$$

$$P(X_2 = R) = 0.8(0.2) + 0.2(0.6) = 0.28.$$

Thus

$$\boldsymbol{\mu}_2 = (0.72, 0.28).$$

k steps use a matrix power

Repeated substitution gives

$$\boldsymbol{\mu}_k = \boldsymbol{\mu}_0 P^k.$$

Entry $\left(P^k\right)_{ij}$ is the probability of moving from state i to state j in k steps.

This is L4′s repeated linear map, now applied to probabilities.

Verify in NumPy

```
import numpy as np

P = np.array([[.8, .2],
              [.4, .6]])
mu0 = np.array([1., 0.])

print(mu0 @ P)                          # [0.8  0.2 ]
print(mu0 @ np.linalg.matrix_power(P, 2))  # [0.72 0.28]
print(P.sum(axis=1))                    # [1. 1.]
```

Question: one matrix step

Q QUESTION

If $\mu_t = (0.25, 0.75)$, what is $P(X_{t+1} = S)$?

A. 0.25

B. 0.40

C. 0.50

D. 0.75

Answer

A ANSWER

Correct: C — 0.50

Why: $0.25(0.8) + 0.75(0.4) = 0.20 + 0.30 = 0.50.$

Probability of an entire path

Chain rule first

For any sequence,

$$p(x_1, ..., x_T) = p(x_1) \prod_{t=2}^{T} p(x_t | x_1, ..., x_{t-1}).$$

This is the probability chain rule; no Markov assumption yet.

Apply the Markov assumption

Replace the full history by the previous state:

$$\boxed{p(x_1, \ldots, x_T) = p(x_1) \prod_{t=2}^{T} P_{x_{t-1}, x_t}.}$$

A path probability is the initial-state probability times the transition probabilities along its arrows.

Worked path: S, R, R, S

Let $P(X_1 = S) = 0.6$ and $P(X_1 = R) = 0.4$.

$$P(S, R, R, S) = P(S)P(R|S)P(R|R)P(S|R).$$

$$= 0.6 \times 0.2 \times 0.6 \times 0.4.$$

$$= 0.0288.$$

Compare two paths

$P(S, S, S, S) = 0.6(0.8)^3 = 0.3072.$

$P(S, R, R, S) = 0.0288.$

Both paths are possible, but the first follows high-probability transitions throughout.

The log path probability is a sum, exactly as in L14 and L24.

Log probability of a path

$$\log p(x_1, ..., x_T) = \log p(x_1) + \sum_{t=2}^{T} \log P_{x_{t-1},x_t}.$$

Negative average log probability is a cross-entropy rate for sequences.

This is how L26 will score a fitted character model.

Sample a chain

```python
rng = np.random.default_rng(4)
state = 0                        # 0=sunny, 1=rainy
path = [state]

for _ in range(12):
    state = rng.choice(2, p=P[state])
    path.append(state)

print(path)
```

Sampling uses the row corresponding to the current state.

A distribution unchanged by one step

Definition

A stationary distribution $\boldsymbol{\pi}$ satisfies

$$\boldsymbol{\pi} = \boldsymbol{\pi} P, \quad \sum_i \pi_i = 1, \quad \pi_i \geq 0.$$

If X_t has distribution $\boldsymbol{\pi}$, then X_{t+1} has the same distribution.

Stationary does not mean the state stops changing; it means the distribution over states is unchanged.

⭐ Solve the weather equations

D DERIVATION

Let $\boldsymbol{\pi} = (\pi_S, \pi_R)$.

$$(\pi_S, \pi_R) = (\pi_S, \pi_R) \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

The sunny coordinate gives

$$\pi_S = 0.8\pi_S + 0.4\pi_R.$$

Therefore

$$0.2\pi_S = 0.4\pi_R \quad \Rightarrow \quad \pi_S = 2\pi_R.$$

⭐ Normalize

DERIVATION

$\pi_S + \pi_R = 1$

and $\pi_S = 2\pi_R$ imply

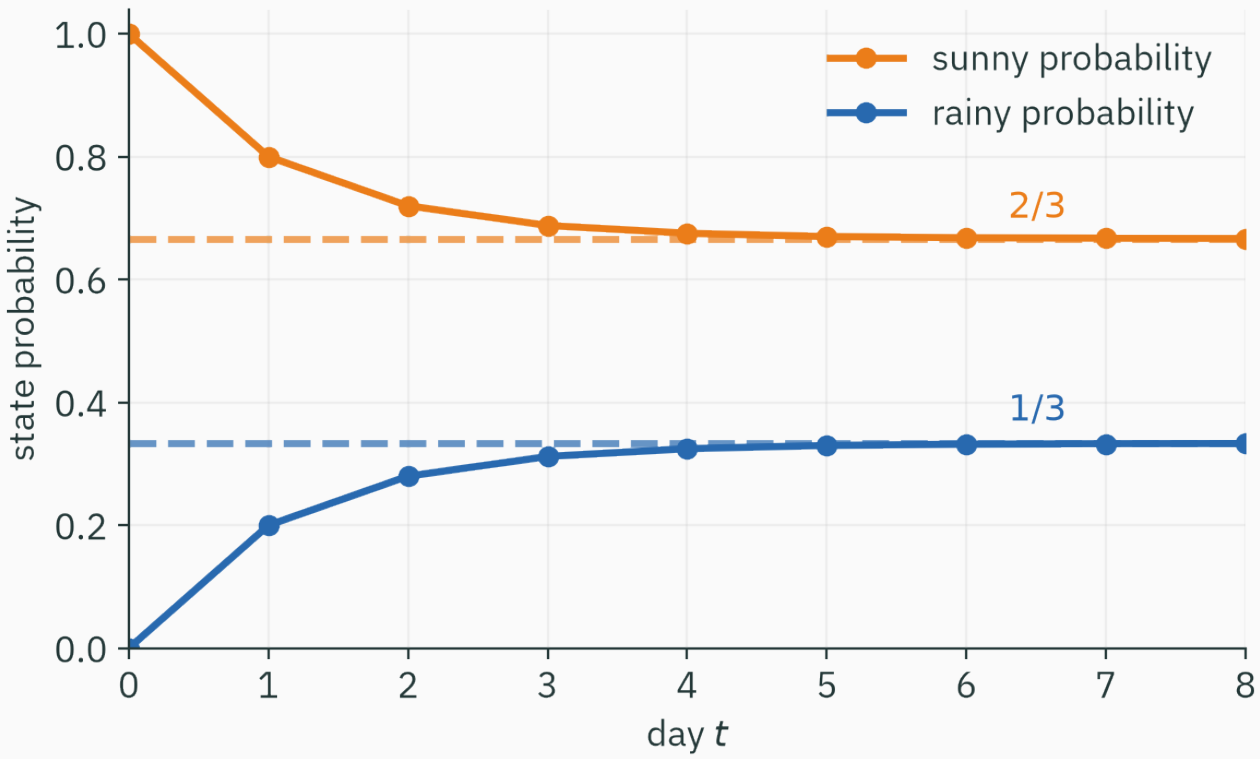
$3\pi_R = 1.$

$$\boxed{\boldsymbol{\pi} = \left(\tfrac{2}{3}, \tfrac{1}{3}\right).}$$

In the long-run balance, sunny is twice as likely as rainy.

Convergence from a sunny start

V VISUAL



The initial condition is gradually forgotten, and the distribution approaches $\left(\frac{2}{3}, \frac{1}{3}\right)$.

Stationarity is an eigenvector equation

$\boldsymbol{\pi} = \boldsymbol{\pi} P$

means $\boldsymbol{\pi}$ is a **left eigenvector** of P with eigenvalue 1.

Equivalently,

$P^\top \boldsymbol{\pi}^\top = \boldsymbol{\pi}^\top.$

This is the eigenvalue-one direction from L5, normalized to sum to one.

Why eigenvalue one exists

Since every row of P sums to one,

$$P\mathbf{1} = \mathbf{1}.$$

Thus $\mathbf{1}$ is a right eigenvector with eigenvalue one.

The stationary distribution is the corresponding left eigenvector. Under suitable conditions it is unique.

Power iteration

Choose any initial distribution and repeat

$$\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t P.$$

This is L5′s power iteration, now constrained to probability vectors.

```
mu = np.array([1., 0.])
for _ in range(30):
    mu = mu @ P
print(mu)  # [0.66666667 0.33333333]
```

The second eigenvalue sets the rate

★ OPTIONAL

The weather matrix has eigenvalues 1 and 0.4.

The non-stationary component shrinks by a factor of 0.4 per step:

$\boldsymbol{\mu}_t - \boldsymbol{\pi} \propto 0.4^t$.

An eigenvalue closer to one would mean slower mixing.

Connectivity and periodicity

Irreducibility

A finite chain is **irreducible** if every state can eventually reach every other state with positive probability.

Then the chain has one communicating class and a unique stationary distribution.

If the chain is reducible, different closed groups can retain memory of the starting group.

Aperiodicity

A state has period d if returns to it can occur only at times sharing greatest common divisor d.

A chain is aperiodic if its states have period one.

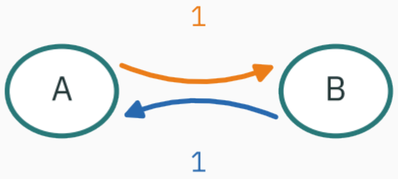
A positive self-loop is a common sufficient reason for aperiodicity.

Two failure patterns

reducible: two closed groups

long-run behaviour depends on the starting group

periodic: alternate forever



[1, 0] and [0, 1] keep swapping

Irreducibility concerns reachability. Aperiodicity concerns synchronized cycles.

Unique stationary does not imply convergence

For the two-state alternator

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

the stationary distribution is $\left(\frac{1}{2}, \frac{1}{2}\right)$.

But starting from $(1, 0)$ gives

$$(1, 0), (0, 1), (1, 0), (0, 1), \ldots$$

The chain is irreducible but periodic, so the distributions do not converge.

A standard convergence statement

For a finite, irreducible, aperiodic Markov chain:

1. a unique stationary distribution exists,
2. every initial distribution converges to it,
3. long-run state frequencies converge to stationary probabilities.

This is the operating theorem behind PageRank and many sampling algorithms.

A random walk on the web

Links define transitions

Imagine a user on a web page choosing one outgoing link uniformly.

For four pages:

1. A links to B and C,
2. B links to C,
3. C links to A,
4. D links to C.

The undamped transition matrix is

$$S = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The random-surfer problem

The link chain may be reducible, periodic, or contain dangling pages with no outgoing links.

PageRank adds teleportation:

$$P = \alpha S + (1-\alpha)\frac{1}{4}\mathbf{1}\mathbf{1}^\top.$$

With probability α, follow a link. With probability $1-\alpha$, jump uniformly to any page.

Why teleportation helps

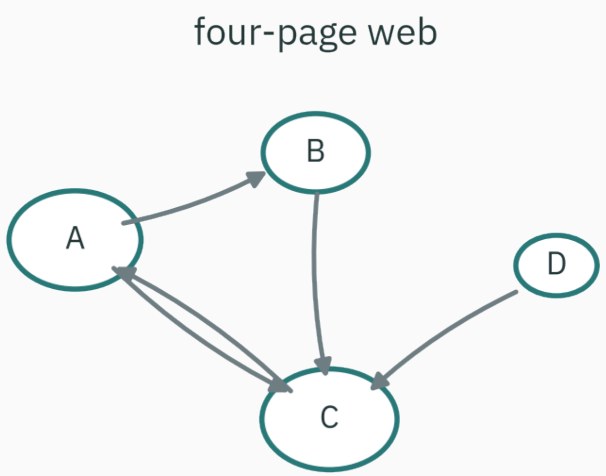
For $0 < \alpha < 1$, every entry of P is positive.

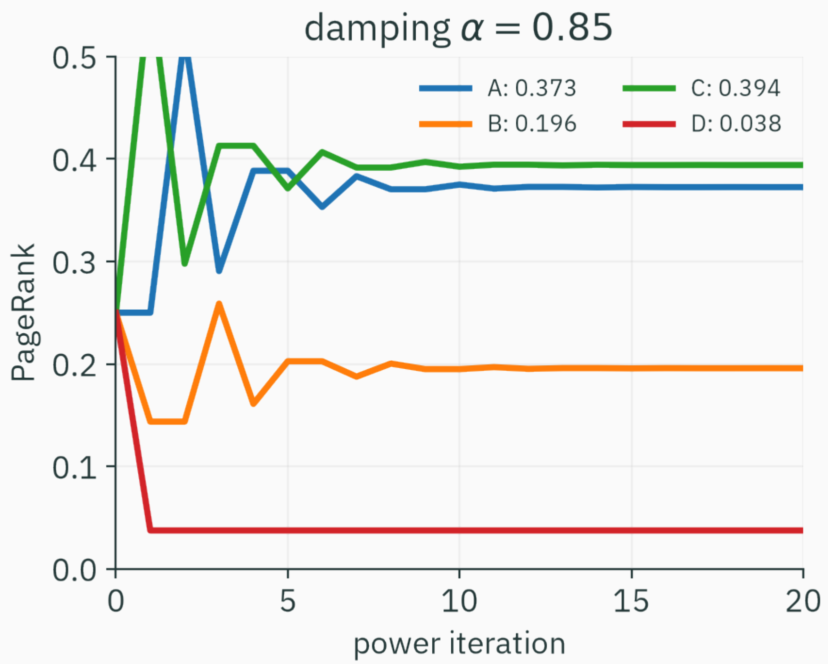
Therefore:

1. every page can reach every other page in one step,
2. every page has a positive self-transition probability,
3. the chain is irreducible and aperiodic.

The stationary distribution is unique and power iteration converges.

Four-page PageRank





With $\alpha = 0.85$, the stationary scores are approximately

$$(0.373, 0.196, 0.394, 0.038)$$

Four-page PageRank

V VISUAL

for (A, B, C, D).

Interpret the scores

Page C receives links from A, B, and D, so its stationary probability is largest.

Page A receives from C, which itself has high rank.

Page D has no incoming web links; it receives only teleportation mass.

PageRank values are visitation probabilities under a specific random-surfer model, not a universal measure of page quality.

Power iteration for PageRank

```
S = np.array([[0, .5, .5, 0],
              [0,  0,  1, 0],
              [1,  0,  0, 0],
              [0,  0,  1, 0]], dtype=float)

alpha = 0.85
P = alpha*S + (1-alpha)*np.ones((4, 4))/4
r = np.ones(4)/4
for _ in range(100):
    r = r @ P
print(r)
```

Sparse computation at web scale

★ OPTIONAL

A web transition matrix is enormous but sparse: each page links to only a small fraction of all pages.

Power iteration needs matrix–vector products, not a dense matrix inverse.

The same principle appeared throughout the course:

1. solve systems without forming inverses,
2. apply structured matrices without storing zeros,
3. exploit repeated local operations.

What the state must remember

A higher-order chain

If the last two states matter, define

$$P(X_{t+1}|X_t, X_{t-1}).$$

This second-order model can be converted to a first-order chain by defining an expanded state

$$Z_t = (X_{t-1}, X_t).$$

The Markov property depends on how the state is defined.

State design is modeling

A useful state should contain enough information to predict the next step for the intended task.

Too small:

1. important dependence remains in the history.

Too large:

1. transition tables become sparse,
2. more parameters must be estimated,
3. data requirements grow quickly.

Time-homogeneous versus changing chains

We assumed one matrix P for every time step:

$$P(X_{t+1} = j | X_t = i) = P_{ij}.$$

Seasonal or intervention-driven systems may require P_t that changes with time.

The matrix framework remains useful, but stationarity may no longer be the right question.

A sequence becomes linear algebra

Summary

A finite Markov chain is a stochastic matrix; its long-run distribution is an eigenvector with eigenvalue one.

1. The Markov assumption keeps only the current state when predicting the next.
2. Sequence probabilities multiply transition probabilities along a path.
3. State distributions evolve as $\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t P$.
4. A stationary distribution satisfies $\boldsymbol{\pi} = \boldsymbol{\pi} P$.
5. Irreducibility gives one communicating class; aperiodicity prevents synchronized oscillation.
6. PageRank is a damped random walk solved by power iteration.

Practice

1. For $P = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$, find the stationary distribution.
2. Starting from state 1, compute the probability of state 2 after two steps.
3. Find the probability of path $(1, 1, 2, 2)$ if $P(X_1 = 1) = 0.6$.
4. Explain whether $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is irreducible and aperiodic.

Practice answers

A ANSWER

1. Solve $0.1\pi_1 = 0.3\pi_2$ and normalize: $\boldsymbol{\pi} = (0.75, 0.25)$.
2. $\left(P^2\right)_{12} = 0.9(0.1) + 0.1(0.7) = 0.16$.
3. $0.6(0.9)(0.1)(0.7) = 0.0378$.
4. It is irreducible because each state reaches the other; it is periodic with period two, so it is not aperiodic.

Next: estimate and sample

So far P was given. Data usually provide only observed transitions.

Next lecture:

1. estimate rows of P by maximum likelihood,
2. smooth counts with a Dirichlet prior,
3. score held-out sequences by cross-entropy,
4. sample characters from a fitted model.

Repeated transition-matrix multiplication turns local rules into long-run behaviour.